

A Small-Gain Approach to Robust Event-Triggered Control of Nonlinear Systems

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Abstract—This paper presents a new approach to event-triggered control for nonlinear uncertain systems by using the notion of input-to-state stability (ISS) and the nonlinear small-gain theorem. The contribution of this paper is threefold. First, it is proved that infinitely fast sampling can be avoided if the system is input-to-state stabilizable with the sampling error as the external input and the corresponding ISS gain is locally Lipschitz. No assumption on the existence of known ISS-Lyapunov functions is made in the discussions. Moreover, the forward completeness problem with event-triggered control is studied systematically by using ISS small-gain arguments. Second, the proposed approach gives rise to a new self-triggered sampling strategy for a class of nonlinear systems subject to external disturbances. If an upper bound of the external disturbance is known, then the closed-loop system can be designed to be robust to the external disturbance, and moreover, the system state globally asymptotically converges to the origin if the external disturbance decays to zero. Third, a new design method is developed for event-triggered control of nonlinear uncertain systems in the strict-feedback form. It is particularly shown that the ISS gain with the sampling error as the input can be designed to satisfy the proposed condition for event-triggered control and self-triggered control.

Index Terms—Cyclic-small-gain theorem, event-triggered control, input-to-state stability (ISS), nonlinear systems, robust control.

I. INTRODUCTION

A. Background

THE study of event-triggered control has recently attracted considerable attention within the systems and control community. A usually considered event-triggered control system can be viewed as a sampled-data system in which data sampling is triggered by external events depending on the real-time system state, and may not be periodic. Compared with the traditional periodic sampling, event-triggered sampling takes into account the system behavior between the sampling time

instants and has been proved to be quite useful in reducing the waste of computation and communication resources in networked control systems. Event-triggered sampling also provides solutions to sampled-data control of nonlinear systems, for which periodic sampling may not work very well.

Significant contributions have been made to the literature of event-triggered control; see, e.g., [1]–[8] and the references therein. Specifically, in [2], [6], impulsive control methods are developed to keep the states of first-order stochastic systems inside certain thresholds. In [9], [10], prediction of the real-time system state between the sampling time instants was employed to generate the control signal, and the prediction is corrected by data-sampling when the difference between the true state and the predicted state is too large. In this paper, we focus on the problems with nonlinear systems. Reference [4] considered the systems which admit controllers to guarantee the robustness with respect to the sampling errors. Then, the event trigger is designed such that the sampling error is bounded by a specific threshold (depending on the real-time system state) for convergence of the system state. Reference [11] proposes a universal formula for event-based stabilization of general nonlinear systems affine in the control by extending Sontag's result for continuous-time stabilization [12]. Reference [13] proposes a Lyapunov condition for tracking control of nonlinear systems. The designs have been extended to distributed networked control [14], [15], output-feedback control and decentralized control [8] and systems with quantized measurements [16], to name a few.

For practical implementation of event-triggered control, infinitely fast sampling should be avoided, that is, the intervals between the sampling time instants should be lower bounded by some positive constant [17]. In the context of event-based control, due to the hybrid nature, the forward completeness of the closed-loop system is a complex issue. This problem has not been studied in details in the literature of event-based nonlinear control, but will be examined in the present paper.

In the process of event-triggered control, the real-time system state should be continuously monitored, which may not be easy in practice. An alternative approach is the self-triggered control, with which the controller computes the control signal as well as the next sampling time instant such that continuous monitoring of system state is not needed [18]. Recent results on self-triggered control can be found in [14], [19]–[24]. When systems are subject to external disturbances, finite-gain $\mathcal{L}_p$ stability [19], [14] can be realized. Reference [20] presents a self-triggered control technique for homogeneous systems and polynomial systems. Polynomial approximations are employed for self-triggered control of nonlinear systems in [22]. In [24],

Manuscript received November 20, 2013; revised June 7, 2014; accepted January 19, 2015. Date of publication January 29, 2015; date of current version July 24, 2015. This work was supported in part by NSF grants DMS-0906659, ECCS-1230040, in part by NSFC grant 61374042, in part by the Fundamental Research Funds for the Central Universities under Grants N130108001 and N140805001 in China, and in part by State Key Laboratory of Intelligent Control and Decision of Complex Systems. Recommended by Associate Editor C. Prieur.

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Digital Object Identifier 10.1109/TAC.2015.2396645

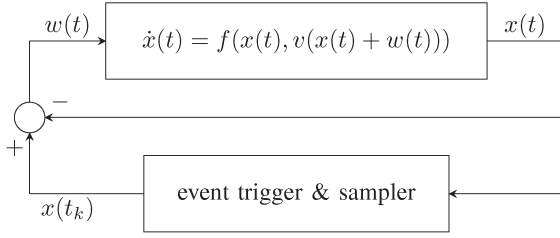


Fig. 1. Event-triggered control system as an interconnection of two subsystems.

self-triggered sampling has been successfully applied to robotic networks. The reader may consult the nice tutorial [25] for the recent developments of event-triggered control and self-triggered control.

The notion of input-to-state stability (ISS), invented by Sontag, is a powerful tool to describe the stability property of nonlinear systems with external inputs [26]. For event-triggered control, ISS has been used to describe the influence of data-sampling to control [4], [20], [21]. In this framework, it is usually assumed that a system is input-to-state stabilizable with the sampling as the external input, and a known ISS-Lyapunov function is also assumed. The basic idea is to find an appropriate event trigger such that the influence of data-sampling is attenuated and the ISS-Lyapunov function is then a Lyapunov function of the closed-loop system augmented with the event-triggered sampling. Usually, this can be realized by designing the event trigger to guarantee that the measurement error caused by data-sampling is always smaller than a well chosen threshold depending on the real-time system state. In the very recent paper [15], the ISS small-gain theorem is applied to guarantee the stability of the overall system, and a parsimonious event triggering mechanism is developed to avoid the Zeno behavior.

B. Contributions of This Paper

By considering an event-triggered control system as an interconnection of the controlled system and the event trigger (as shown in Fig. 1), in this paper, we develop a new approach to event-triggered control of nonlinear systems based on the small-gain theorem of [27] and its extensions. Specifically, we propose an ISS gain condition and correspondingly an event triggering mechanism to avoid infinitely fast sampling, and at the same time, to achieve asymptotic stabilization. The forward completeness issue with event-triggered control is studied systematically by using ISS small-gain arguments. As an extension of the event-triggered mechanism, a self-triggered sampling strategy is also proposed.

This paper also presents a self-triggered sampling strategy for nonlinear systems subject to external disturbances. As is well-known, external disturbances may lead to infinitely fast sampling for the systems which work well when there is no external disturbance [8]. This problem can be solved by introducing an ϵ term to the event trigger such that the threshold value for the norm of the sampling error is always strictly positive. The price paid for this is the loss of asymptotic convergence, even if the external disturbance decays to zero.

With the proposed small-gain approach, if an upper bound of the external disturbance is known, then the closed-loop system can be designed to be robust to the external disturbance, and the system state asymptotically converges to the origin if the external disturbance converges to zero.

Input-to-state stabilization in the presence of measurement errors plays a critical role in the designs for event-triggered control and self-triggered control in this paper. This could be easily realizable for stabilizable linear systems. However, small converging measurement error may cause the performance of a nonlinear control system to deteriorate, even if the system with no measurement error is asymptotically stable [28]. In this paper, we also contribute a novel design for event-triggered control of the benchmark nonlinear systems in the strict-feedback form [29]. Motivated by the set-valued map design in [30], we employ set-valued maps to cover the influence of the sampling error, and transform the closed-loop system into a network of ISS subsystems. With the cyclic-small-gain theorem [31], [32], ISS of the closed-loop system with the sampling error as the input is guaranteed, and the influence of the sampling error is explicitly evaluated. More importantly, it is shown that a nonlinear control law can be designed to satisfy the ISS gain condition to avoid infinitely fast sampling as long as the system has an equilibrium at the origin and the function of the system dynamics is locally Lipschitz. It should be noted that this design does not require the accurate knowledge of the system dynamics.

The rest of the paper is organized as follows. Section II presents the notations and reviews the concept of ISS and its small-gain results that will be used for the designs in this paper. In Section III, we propose an ISS gain condition for even-triggered control and self-triggered control of nonlinear systems. A new self-triggered control design is proposed in Section IV to deal with the problems caused by external disturbances. In Section V, we focus on nonlinear systems in the strict-feedback form and develop feedback control laws to fulfill the requirements for event-triggered control and self-triggered control. Section VI gives an example to show the basic design procedure. In Section VII, we present some concluding remarks and outline some related topics for future research.

II. PRELIMINARIES

In this section, we present some basic notations and review the concept of ISS and the nonlinear small-gain results that will be used in the rest of this paper.

A continuous function $\alpha : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is said to be positive definite if $\alpha(0) = 0$ and $\alpha(s) > 0$ for $s > 0$. A continuous function $\alpha : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is said to be a class $\mathcal{K}$ function, denoted by $\alpha \in \mathcal{K}$, if it is strictly increasing and $\alpha(0) = 0$; it is said to be a class $\mathcal{K}_\infty$ function, denoted by $\alpha \in \mathcal{K}_\infty$, if it is a class $\mathcal{K}$ function and satisfies $\alpha(s) \rightarrow \infty$ as $s \rightarrow \infty$. In this paper, $\gamma_1 \circ \gamma_2$ represents the composition function of γ_1 and γ_2 . For $\gamma_1, \gamma_2 \in \mathcal{K}$, $\gamma_1 \circ \gamma_2 < \text{Id}$ means $\gamma_1(\gamma_2(s)) < s$ for all $s > 0$. A continuous function $\beta : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is said to be a class $\mathcal{KL}$ function, denoted by $\beta \in \mathcal{KL}$, if, for each fixed $t \in \mathbb{R}_+$, $\beta(\cdot, t)$ is a class $\mathcal{K}$ function and, for each fixed $s \in \mathbb{R}_+$, $\beta(s, \cdot)$ is a decreasing function satisfying $\lim_{t \rightarrow \infty} \beta(s, t) = 0$.

For systems with external inputs, the notion of ISS invented by Sontag has been proved to be powerful in evaluating the influence of the external inputs.

Consider system

$$\dot{x} = f(x, u) \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ represents the input, and $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a locally Lipschitz function and satisfies $f(0, 0) = 0$. Assume that u is measurable and locally essentially bounded.

In [33], the original definition of ISS is given in the “plus”-form. For convenience of discussions, we mainly use the equivalent “max”-form definition.

Definition 1: System (1) is said to be ISS if there exist a $\beta \in \mathcal{KL}$ and a $\gamma \in \mathcal{K}$ such that for any initial state $x(0) = x_0$ and any measurable and locally essentially bounded u , the solution $x(t)$ satisfies

$$|x(t)| \leq \max \{ \beta(|x_0|, t), \gamma(\|u_t\|_\infty) \} \quad (2)$$

for all $t \geq 0$, where u_t is the truncated function of u at t . Here, γ is called ISS gain of the system. By causality, we can replace the L_∞ -norm of u_t by $\|u\|_\infty$.

The small-gain theorem developed in [27] has been proved to be very useful in the analysis and controller design of interconnected nonlinear systems. In this paper, the systems will be considered as interconnected systems in the event-triggered control designs. Consider system

$$\dot{x}_i = f_i(x, u_i), \quad i = 1, 2 \quad (3)$$

where $x = [x_1^T, x_2^T]^T$ with $x_1 \in \mathbb{R}^{n_1}$ and $x_2 \in \mathbb{R}^{n_2}$ is the state, $u_1 \in \mathbb{R}^{m_1}$ and $u_2 \in \mathbb{R}^{m_2}$ are external inputs, $f_1 : \mathbb{R}^{n_1+n_2} \times \mathbb{R}^{m_1} \rightarrow \mathbb{R}^{n_1}$ and $f_2 : \mathbb{R}^{n_1+n_2} \times \mathbb{R}^{m_2} \rightarrow \mathbb{R}^{n_2}$ are locally Lipschitz functions satisfying $f_1(0, 0) = 0$ and $f_2(0, 0) = 0$. For convenience of notations, denote $u = [u_1^T, u_2^T]^T$. By considering u as a function of time, assume that u is measurable and locally essentially bounded.

For $i = 1, 2$, assume that each x_i -subsystem is ISS with x_{3-i} and u_i as inputs. Specifically, for each $i = 1, 2$, there exist $\beta_i \in \mathcal{KL}$ and $\gamma_{i(3-i)}, \gamma_i^u \in \mathcal{K}$ such that for any initial state $x_i(0) = x_{i0}$ and any measurable and locally essentially bounded inputs x_{3-i}, u_i , the solution $x_i(t)$ satisfies

$$|x_i(t)| \leq \max \{ \beta_i(|x_{i0}|, t), \gamma_{i(3-i)}(\|x_{3-i}\|_\infty), \gamma_i^u(\|u_i\|_\infty) \} \quad (4)$$

for all $t \geq 0$.

Reference [27] considers systems in a more general form with the subsystems interconnected by their outputs and assumed to be input-to-output stable (IOS). Here, we just review the special case in which the subsystems are ISS and are directly interconnected by their states.

Theorem 1: Consider the interconnected system composed of two subsystems in the form of (3) satisfying (4). The interconnected system is ISS with u as the input if the small-gain condition

$$\gamma_{12} \circ \gamma_{21} < \text{Id} \quad (5)$$

is satisfied.

Remark 1: It is obvious that Theorem 1 still holds if one of the subsystems, say the x_2 -subsystem, satisfies (4) with $\beta_2 = 0$. This is the case when the x_2 -subsystem is memoryless. Also, due to causality, if (4) holds for all $t \in [0, T_{\max})$ with $0 < T_{\max} \leq \infty$, then with (5) satisfied, the solution $x(t)$ of interconnected system (3) satisfies (2) for all $t \in [0, T_{\max})$.

Recently, the small-gain result in [27] has been generalized to address problems arising from large-scale systems. With the new result, the IOS property of a large-scale system composed of IOS subsystems can be tested by checking the composition of the IOS gains along every simple cycle of the system interconnection structure [31]. Here, due to space limitation, we only review the special case in which the large-scale system is composed of ISS subsystems and each subsystem has an ISS-Lyapunov function [32]. Note that a nonlinear system in the form of (1) is ISS if and only if it admits an ISS-Lyapunov function [34].

Consider system

$$\dot{x}_i = f_i(x, u_i), \quad i = 1, \dots, N \quad (6)$$

where $x = [x_1^T, \dots, x_N^T]^T$ with $x_i \in \mathbb{R}^{n_i}$ is the state, $u_i \in \mathbb{R}^{m_i}$ represents the external input of the x_i -subsystem, and each $f_i : \mathbb{R}^{n+m_i} \rightarrow \mathbb{R}^{n_i}$ with $n = \sum_{j=1}^N n_j$ is a locally Lipschitz function satisfying $f_i(0, 0) = 0$. The external input $u = [u_1^T, \dots, u_N^T]^T$ is a measurable and locally essentially bounded function from $\mathbb{R}_+$ to $\mathbb{R}^m$ with $m = \sum_{i=1}^N m_i$. Denote $f(x, u) = [f_1^T(x, u_1), \dots, f_N^T(x, u_N)]^T$.

Assume that for $i = 1, \dots, N$, each x_i -subsystem admits a continuously differentiable ISS-Lyapunov function $V_i : \mathbb{R}^{n_i} \rightarrow \mathbb{R}_+$ satisfying

1) there exist $\underline{\alpha}_i, \bar{\alpha}_i \in \mathcal{K}_\infty$ such that

$$\underline{\alpha}_i(|x_i|) \leq V_i(x_i) \leq \bar{\alpha}_i(|x_i|), \quad \forall x_i; \quad (7)$$

2) there exist $\gamma_{ij} \in \mathcal{K} \cup \{0\}$ ($j = 1, \dots, N, j \neq i$) and $\gamma_{ui} \in \mathcal{K} \cup \{0\}$ such that

$$\begin{aligned} V_i(x_i) &\geq \max_{j \neq i} \{ \gamma_{ij}(V_j(x_j)), \gamma_{ui}(\|u_i\|) \} \\ &\Rightarrow \nabla V_i(x_i) f_i(x, u_i) \leq -\alpha_i(V_i(x_i)), \quad \forall x, u_i \end{aligned} \quad (8)$$

where α_i is a positive definite function.

Theorem 2 ([32]): Consider the large-scale system (6) with each x_i -subsystem admitting a continuously differentiable ISS-Lyapunov function V_i satisfying (7), (8). Then, it is ISS with x as the state and u as the input if for every simple cycle $(i_1, i_2, \dots, i_r, i_1)$ in the gain digraph

$$\gamma_{i_1 i_2} \circ \gamma_{i_2 i_3} \circ \dots \circ \gamma_{i_r i_1} < \text{Id} \quad (9)$$

where $r = 2, \dots, N$ and $1 \leq i_j \leq N, i_j \neq i_{j'}$ if $j \neq j'$. Moreover, an ISS-Lyapunov function can be constructed for the large-scale system (6)

$$V(x) = \max_{i=1, \dots, N} \{ \sigma_i(V_i(x)) \} \quad (10)$$

where σ_i are appropriately chosen class $\mathcal{K}_\infty$ functions being locally Lipschitz on $(0, \infty)$.

In this paper, the ISS small-gain result given by Theorem 1 will be used in Section III to develop a new ISS gain condition to avoid infinitely fast sampling for event-triggered control of nonlinear systems. Theorem 1 will also be used for the designs to handle external disturbances in Section IV. In Section V, we will consider nonlinear uncertain systems in the strict-feedback form. With the control design, we will design a controller to transform the closed-loop system into a large-scale system composed of ISS subsystems, and use Theorem 2 to guarantee the ISS of the closed-loop system. By means of (10), an ISS-Lyapunov function will be constructed to evaluate the influence of the sampling error.

III. AN ISS GAIN CONDITION FOR EVENT-TRIGGERED CONTROL

An event-triggered state-feedback control system is generally in the following form:

$$\dot{x}(t) = f(x(t), u(t)) \quad (11)$$

$$u(t) = v(x(t_k)), \quad t \in [t_k, t_{k+1}), k \in \mathbb{S} \subseteq \mathbb{Z}_+ \quad (12)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input, $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is a locally Lipschitz function representing system dynamics, $v: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a locally Lipschitz function representing the control law. It is assumed that $f(0, v(0)) = 0$. The time sequence $\{t_k\}_{k \in \mathbb{S}}$ is determined online based on the measurement of the real-time system state. Suppose that $x(t)$ is right maximally defined for all $t \in [0, T_{\max})$ with $0 < T_{\max} \leq \infty$. With respect to the possible finite-time accumulation of t_k and finite-time divergence of $x(t)$, we consider three cases:

- 1) $\mathbb{S} = \mathbb{Z}_+$ and $\lim_{k \rightarrow \infty} t_k < \infty$, which means Zeno behavior [35].
- 2) $\mathbb{S} = \mathbb{Z}_+$ and $\lim_{k \rightarrow \infty} t_k = \infty$. In this case, $x(t)$ is defined on $[0, \infty)$.
- 3) $\mathbb{S}$ is a finite set $\{0, \dots, k^*\}$ with $k^* \in \mathbb{Z}_+$, i.e., there is a finite number of sampling time instants. In this case, $t_{k^*} < T_{\max}$ and we set $t_{k^*+1} = T_{\max}$ for convenience of discussions.

It should be noted that, in any case, $x(t)$ is defined for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. With an appropriate event trigger design, we will show that $\inf_{k \in \mathbb{S}} \{t_{k+1} - t_k\} > 0$, which means that Case 1) is impossible. Also, by means of small-gain arguments, we will prove that $T_{\max} = \infty$ for Case 3).

Define

$$w(t) = x(t_k) - x(t), \quad t \in [t_k, t_{k+1}), k \in \mathbb{S} \quad (13)$$

as the sampling error, and rewrite

$$u(t) = v(x(t) + w(t)). \quad (14)$$

By substituting (12) into (11), we have

$$\dot{x}(t) = f(x(t), v(x(t_k))) =: \bar{f}(x(t), x(t_k)). \quad (15)$$

Then, by using (14), we have

$$\dot{x}(t) = \bar{f}(x(t), x(t) + w(t)). \quad (16)$$

If $w(t)$ is not adjustable, then the event-triggered control problem is reduced to the measurement feedback control problem. The basic idea of event-triggered control is to adjust $w(t)$ online with an appropriate data-sampling strategy, to realize asymptotic convergence of $x(t)$, if possible. From this point of view, the structure of the closed-loop system can be represented with the block diagram shown in Fig. 1.

By taking the advantage of the interconnection structure, we develop a nonlinear small-gain approach to event-triggered control of nonlinear systems.

Assumption 1: System (16) is ISS with w as the input, that is, there exist $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that for any initial state $x(0)$ and any measurable and locally essentially bounded w , it holds that

$$|x(t)| \leq \max\{\beta(|x(0)|, t), \gamma(\|w\|_\infty)\} \quad (17)$$

for all $t \geq 0$.

According to Theorem 1, under Assumption 1, if the event trigger is designed such that $|w(t)| \leq \rho(|x(t)|)$ for all $t \geq 0$ with $\rho \in \mathcal{K}$ satisfying

$$\rho \circ \gamma < \text{Id}, \quad (18)$$

then $x(t)$ asymptotically converges to the origin. Based on this idea, the event trigger considered in this paper can be defined as follows: if $x(t_k) \neq 0$, then

$$t_{k+1} = \inf\{t > t_k : \rho(|x(t)|) - |x(t) - x(t_k)| = 0\}. \quad (19)$$

The data sampling event is not triggered if for some specific $k^* \in \mathbb{Z}_+$, $x(t_{k^*}) = 0$ or $\{t > t_k : \rho(|x(t)|) - |x(t) - x(t_k)| = 0\} = \emptyset$. Note that, under the assumption of $f(0, v(0)) = 0$, if $x(t_{k^*}) = 0$, then $u(t) = v(x(t_{k^*})) = 0$ keeps the system state at the origin for all $t \in [t_{k^*}, \infty)$.

With the event trigger (19), given t_k and $x(t_k) \neq 0$, t_{k+1} is the first time instant after t_k such that $\rho(|x(t_{k+1})|) - |x(t_{k+1}) - x(t_k)| = 0$. Since $\rho(|x(t_k)|) - |x(t_k) - x(t_k)| = \rho(|x(t_k)|) > 0$ for any $x(t_k) \neq 0$ and $x(t)$ is continuous on $[0, \infty)$, we have $\rho(|x(t)|) - |x(t) - x(t_k)| > 0$ for all $t \in [t_k, t_{k+1})$, $k \in \mathbb{S}$. By using the definition of $w(t)$ in (13), we have

$$|w(t)| \leq \rho(|x(t)|) \quad (20)$$

for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$.

Theorem 3 presents a condition on the ISS gain γ to find a ρ for the event trigger (19) such that infinitely fast sampling is avoided by $\inf_{k \in \mathbb{S}} \{t_{k+1} - t_k\} > 0$, and moreover, $x(t)$ is defined for all $t \in [0, \infty)$ and asymptotically converges to the origin.

Theorem 3: Consider the event-triggered control system (16) with locally Lipschitz $\bar{f}$ satisfying $\bar{f}(0, 0) = 0$ and w defined in (13). If Assumption 1 is satisfied with a locally Lipschitz γ , then one can find a $\rho \in \mathcal{K}_\infty$ such that

- ρ satisfies (18), and
- ρ^{-1} is locally Lipschitz.

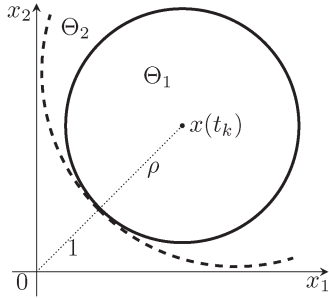


Fig. 2. An illustration of $\Theta_1(x(t_k)) \subseteq \Theta_2(x(t_k))$.

Moreover, with the sampling time instants triggered by (19), it can always be guaranteed that

$$\inf_{k \in \mathbb{S}} \{t_{k+1} - t_k\} > 0 \quad (21)$$

and, for any specific initial state $x(0)$, the system state $x(t)$ satisfies

$$|x(t)| \leq \check{\beta}(|x(0)|, t) \quad (22)$$

with $\check{\beta} \in \mathcal{KL}$, for all $t \geq 0$.

Proof: Using the nonlinear small-gain theorem (Theorem 1), with (17) and (20) satisfied, there exists a $\check{\beta} \in \mathcal{KL}$ such that (22) holds for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. We now prove (21), and then show $\bigcup_{k \in \mathbb{S}} [t_k, t_{k+1}) = [0, \infty)$.

With a locally Lipschitz $\gamma \in \mathcal{K}$, one can always find a locally Lipschitz $\bar{\gamma} \in \mathcal{K}_\infty$ such that $\bar{\gamma} > \gamma$. By choosing $\rho = \bar{\gamma}^{-1}$, we have $\rho \circ \gamma = \bar{\gamma} \circ \gamma < \bar{\gamma} \circ \bar{\gamma}^{-1} < \text{Id}$, and $\rho^{-1} = \bar{\gamma}$ is locally Lipschitz. Along each trajectory of the closed-loop system, for each $k \in \mathbb{S}$ with state $x(t_k)$ at time instant t_k , define

$$\Theta_1(x(t_k)) = \{x \in \mathbb{R}^n : |x - x(t_k)| \leq \rho \circ (\text{Id} + \rho)^{-1}(|x(t_k)|)\} \quad (23)$$

$$\Theta_2(x(t_k)) = \{x \in \mathbb{R}^n : |x - x(t_k)| \leq \rho(|x|)\}. \quad (24)$$

Then, the lower bound of $t_{k+1} - t_k$ can be considered as the minimum time needed for $x(t)$ starting at $x(t_k)$ to go outside $\Theta_2(x(t_k))$. By directly using Lemma A.1, it can be proved that $\Theta_1(x(t_k)) \subseteq \Theta_2(x(t_k))$. An illustration with $x = [x_1, x_2]^T \in \mathbb{R}^2$ is given in Fig. 2. Now we estimate the minimum time needed for $x(t)$ starting at $x(t_k)$ to go outside $\Theta_1(x(t_k))$.

Given a $\rho \in \mathcal{K}_\infty$ such that ρ^{-1} is locally Lipschitz, it can be proved that $\rho^{-1} + \text{Id}$ is locally Lipschitz and there exists a continuous, positive function $\check{\rho} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that

$$(\rho^{-1} + \text{Id})(s) \leq \check{\rho}(s)s =: \hat{\rho}(s) \quad (25)$$

for $s \in \mathbb{R}_+$. Note that $\check{\rho}(s)s = \hat{\rho}(s)$ implies $s = (\check{\rho} \circ \hat{\rho}^{-1}(s))\hat{\rho}^{-1}(s)$. We have

$$\hat{\rho}^{-1}(s) = \frac{s}{\check{\rho} \circ \hat{\rho}^{-1}(s)} =: \bar{\rho}(s). \quad (26)$$

Here, $\bar{\rho} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a continuous and positive function as $\check{\rho}$ is continuous and positive. Then, by using (25) and (26), we have $\rho \circ (\text{Id} + \rho)^{-1}(s) = (\rho^{-1} + \text{Id})^{-1}(s) \geq \hat{\rho}^{-1}(s) = \bar{\rho}(s)s$. This means, if

$$|x - x(t_k)| \leq \bar{\rho}(|x(t_k)|)|x(t_k)|, \quad (27)$$

then $x \in \Theta_1(x(t_k))$.

Note that for the locally Lipschitz $\bar{f}$ as defined in (16), there exists a continuous and positive function $L_{\bar{f}}$ such that

$$\begin{aligned} |\bar{f}(x, x(t_k))| &= |\bar{f}(x - x(t_k) + x(t_k), x(t_k))| \\ &\leq L_{\bar{f}} \left(|x^T - x^T(t_k), x^T(t_k)|^T \right) \\ &\quad \times |x^T - x^T(t_k), x^T(t_k)|^T. \end{aligned} \quad (28)$$

If moreover $|x - x(t_k)| \leq \bar{\rho}(|x(t_k)|)|x(t_k)|$, then there exists a continuous and positive function $\bar{L}$ such that

$$|\bar{f}(x, v(x(t_k)))| \leq \bar{L}(|x(t_k)|)|x(t_k)|. \quad (29)$$

Thus, the minimum time $T_k^{\min}$ needed for $x(t)$ starting at $x(t_k)$ to go outside the region $\Theta_1(x(t_k))$ can be estimated by

$$\begin{aligned} T_k^{\min} &\geq \min \left\{ \frac{\bar{\rho}(|x(t_k)|)|x(t_k)|}{\bar{L}(|x(t_k)|)|x(t_k)|}, T_{\max} \right\} \\ &= \min \left\{ \frac{\bar{\rho}(|x(t_k)|)}{\bar{L}(|x(t_k)|)}, T_{\max} \right\} \end{aligned} \quad (30)$$

which is well defined and strictly larger than zero for any $x(t_k) \neq 0$. Since $\Theta_1(x(t_k)) \subseteq \Theta_2(x(t_k))$ and $x(t)$ is continuous on the time-line, the minimum interval needed for the state starting at $x(t_k)$ to go outside $\Theta_2(x(t_k))$ is not less than $T_k^{\min}$. Then, by using (22) and (30), we have

$$T_k^{\min} \geq \min \left\{ \min \left\{ \frac{\bar{\rho}(|x|)}{\bar{L}(|x|)} : |x| \leq \check{\beta}(|x(0)|, 0) \right\}, T_{\max} \right\} \quad (31)$$

for all $k \in \mathbb{S}$. Since the right-hand side of (31) is strictly positive, property (21) is proved.

Now, we consider the above-mentioned three cases: (1) $\mathbb{S} = \mathbb{Z}_+$ and $\lim_{k \rightarrow \infty} t_k < \infty$; (2) $\mathbb{S} = \mathbb{Z}_+$ and $\lim_{k \rightarrow \infty} t_k = \infty$; (3) $\mathbb{S} = \{0, \dots, k^*\}$ with $k^* \in \mathbb{Z}_+$.

a) Suppose that Case (1) happens. Then, we have $\inf_{k \in \mathbb{S}} \{t_{k+1} - t_k\} = 0$, which contradicts with (21). Thus, Case (1) is impossible.

b) Case (2), we have $\bigcup_{k \in \mathbb{S}} [t_k, t_{k+1}) = [0, \infty)$, which means that (22) holds for all $t \in [0, \infty)$.

c) In Case (3), since $t_{k^*+1} = T_{\max}$, we have $\bigcup_{k \in \mathbb{S}} [t_k, t_{k+1}) = [0, T_{\max})$, and thus (22) holds for all $t \in [0, T_{\max})$. By the continuation of solutions, this implies that $x(t)$ is defined for all $t \in [0, \infty)$, i.e., $T_{\max} = \infty$.

This ends the proof of Theorem 3. $\blacksquare$

Remark 2: Reference [4] assumes the existence of an ISS-Lyapunov function $V : \mathbb{R}^n \rightarrow \mathbb{R}_+$ for the system composed of (11) and (14) with the sampling error w as the input, i.e., $\nabla V(x)f(x, v(x+w)) \leq -\alpha'(|x|) + \gamma'(|w|)$ with $\alpha' \in \mathcal{K}_\infty$ and $\gamma' \in \mathcal{K}$. To avoid the Zeno behavior, it is assumed that both $(\alpha')^{-1}$ and γ' are locally Lipschitz. However, this assumption is not easily checkable even for linear systems. For linear systems with quadratic Lyapunov functions, the α' would be in the form of $\alpha'(s) = ks^2$ with constant $k > 0$, and $(\alpha')^{-1}$ is not locally Lipschitz. With the small-gain approach, the Zeno behavior can be avoided as long as the ISS gain is locally Lipschitz. Our result can be readily applied to linear systems, as discussed in Example 3.1.

Remark 3: For the digital implementation with numerical approximations, it might be more flexible to generalize the event trigger (19) as $t_{k+1} \leq \inf\{t > t_k : \rho(|x(t)|) - |x(t) - x(t_k)| = 0\}$, which still guarantees (20). In this case, the inter-sample periods should be carefully designed to guarantee the existence of a lower bound.

Also, the small-gain approach naturally leads to a self-triggered sampling strategy. Specifically, based on property (30) in the proof of Theorem 3, we may consider the sampling time instants to be generated by

$$t_{k+1} = \frac{\bar{\rho}(|x(t_k)|)}{\bar{L}(|x(t_k)|)} + t_k \quad \text{for } k, k+1 \in \mathbb{S} \subseteq \mathbb{Z}_+. \quad (32)$$

Note that (32) works only when $x(t_k)$ is defined, i.e., $t_k < T_{\max}$. Clearly, the self-triggered sampling mechanism does not continuously monitor the trajectory of $x(t)$.

Theorem 4: Consider system (16) with locally Lipschitz $\bar{f}$ satisfying $\bar{f}(0, 0) = 0$ and w defined in (13). If Assumption 1 is satisfied with a locally Lipschitz γ , then there exist continuous and positive functions $\bar{\rho}, \bar{L} : \mathbb{R}_+ \rightarrow \mathbb{R}_+ \setminus \{0\}$ such that with the self-triggered sampling strategy (32), the system state satisfies (22) for all $t \geq 0$.

Sketch of the Proof: Following the proof of Theorem 3, property (30) still holds for the self-triggered control system. The self-triggered sampling strategy (32) guarantees that $x(t) \in \Theta_2(x(t_k))$ for all $t \in [t_k, t_{k+1})$, $k \in \mathbb{S}$. Then, it is guaranteed that $|x(t) - x(t_k)| \leq \rho(|x(t)|)$, and thus (20) holds for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. Property (22) can then be proved for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$ by directly using Theorem 1. The convergence for all $t \geq 0$ can be still proved by considering the three cases as for the proof of Theorem 3. But it should be noted that Case (3) is also impossible for self-triggered control due to the boundedness of $t_{k+1} - t_k$ according to the self-triggered sampling strategy (32). ■

Example 3.1: Assumption 1 can be readily satisfied by designing a linear controller for the linear time-invariant system $\dot{x} = Ax + Bu$ with $x \in \mathbb{R}^n$ as the state and $u \in \mathbb{R}^m$ as the control input, if the system is controllable. One can find a K such that $A - BK$ is Hurwitz and design $u = -K(x + w)$ with w being the sampling error. Then, $\dot{x} = Ax - BK(x + w) = (A - BK)x - BKw$. With initial state $x(0)$, the solution of the system is $x(t) = e^{(A-BK)t}x(0) - \int_0^t e^{(A-BK)(t-\tau)}BKw(\tau)d\tau$ for $t \geq 0$. It can be verified that $x(t)$ satisfies property (17) with $\beta(s, t) = (1 + 1/\delta)|e^{(A-BK)t}|s$ and $\gamma(s) = (1 + \delta)(\int_0^\infty |e^{(A-BK)\tau}|BK|d\tau)s$, where δ can be selected as any positive constant. Then, $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$. Moreover, γ is locally Lipschitz.

IV. ROBUSTNESS TO EXTERNAL DISTURBANCES

Theorems 3 and 4 do not address the presence of external disturbances. In this section, we consider a class of systems with external disturbances taking the general form

$$\dot{x}(t) = f(x(t), u(t), d(t)) \quad (33)$$

where $d(t) \in \mathbb{R}^{n_d}$ represents the external disturbances, and the other variables are defined as in (11). It is assumed that d is measurable and locally essentially bounded.

By substituting (12) into (33), we have

$$\dot{x}(t) = f(x(t), v(x(t_k)), d(t)) =: \bar{f}(x(t), x(t_k), d(t)). \quad (34)$$

With w defined in (13) as the sampling error, we have

$$\dot{x}(t) = \bar{f}(x(t), x(t) + w(t), d(t)). \quad (35)$$

Corresponding to Assumption 1 for the disturbance-free case, we make the following assumption for system (35).

Assumption 2: System (35) is ISS with w and d as the inputs, that is, there exist $\beta \in \mathcal{KL}$ and $\gamma, \gamma^d \in \mathcal{K}$ such that for any initial state $x(0)$ and any measurable and locally essentially bounded w and d , it holds that

$$|x(t)| \leq \max\{\beta(|x(0)|, t), \gamma(\|w\|_\infty), \gamma^d(\|d\|_\infty)\} \quad (36)$$

for all $t \geq 0$.

Suppose that (20) is guaranteed by the event trigger with $\rho \in \mathcal{K}$ satisfying $\rho \circ \gamma < \text{Id}$. Then, under Assumption 2, by directly using Theorem 1, we can prove that

$$|x(t)| \leq \max\{\check{\beta}(|x(0)|, t), \check{\gamma}^d(\|d\|_\infty)\} \quad (37)$$

with $\check{\beta} \in \mathcal{KL}$ and $\check{\gamma}^d \in \mathcal{K}$. As x converges to the origin, the upper bound of $|w(t)| = |x(t_k) - x(t)|$ converges to zero according to (20). However, due to the presence of the external disturbance d , the function of system dynamics $f(x(t), v(x(t) + w(t)), d(t))$ may not converge to zero as x converges to the origin. This means that the inter-sample period $t_{k+1} - t_k$ could be arbitrarily small.

A. Event-Triggered Sampling With ϵ Modification

Inspired by the recent result [8], we modify the event trigger (19) as

$$t_{k+1} = \inf\{t > t_k : \max\{\rho(|x(t)|), \epsilon\} - |x(t) - x(t_k)| = 0\} \quad (38)$$

with $\rho \in \mathcal{K}$ satisfying $\rho \circ \gamma < \text{Id}$ and constant $\epsilon > 0$. The modified event trigger guarantees

$$|x(t) - x(t_k)| < \max\{\rho(|x(t)|), \epsilon\} \quad (39)$$

for all $t \in [t_k, t_{k+1})$, $k \in \mathbb{S}$. With Theorem 1, there exist $\check{\beta} \in \mathcal{KL}$ and $\check{\gamma}, \check{\gamma}^d \in \mathcal{K}$ such that

$$|x(t)| \leq \max\{\check{\beta}(|x(0)|, t), \check{\gamma}(\epsilon), \check{\gamma}^d(\|d\|_\infty)\} \quad (40)$$

for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. Following a similar reasoning as for the proof of Theorem 3, we can prove $\bigcup_{k \in \mathbb{S}} [t_k, t_{k+1}) = [0, \infty)$. It should be noted that, with $\epsilon > 0$, the function ρ^{-1} is no longer required to be locally Lipschitz. This result is summarized by Theorem 5 without proof.

Theorem 5: Consider the event-triggered control system (35) with locally Lipschitz $\bar{f}$ and w defined in (13). If Assumption 2 is satisfied, with the sampling time instants triggered by (38), for any specific initial state $x(0)$, the system state $x(t)$ satisfies (40) for all $t \geq 0$, with $\check{\beta} \in \mathcal{KL}$ and $\check{\gamma}, \check{\gamma}^d \in \mathcal{K}$, and the inter-sample periods are lower bounded by a positive constant.

For such event-triggered control system, even if $d \equiv 0$, only practical convergence can be guaranteed, that is, $x(t)$ can only be guaranteed to converge to within a neighborhood of the origin $\{x \in \mathbb{R}^n : |x| \leq \check{\gamma}(\epsilon)\}$, where ϵ can be made arbitrarily small. In the next subsection, we present a self-triggered sampling mechanism to overcome this obstacle, under the assumption of an *a priori* known upper bound of $\|d\|_\infty$.

B. Self-Triggered Sampling

In this subsection, we show that if an upper bound of $\|d\|_\infty$ is known *a priori*, then we can design a self-triggered sampling mechanism such that $x(t)$ is practically steered to within a neighborhood of the origin with size depending solely on $\|d\|_\infty$. Moreover, if $d(t)$ converges to zero, then $x(t)$ asymptotically converges to the origin.

Assumption 3: There is a known constant $B^d \geq 0$ such that

$$\|d\|_\infty \leq B^d. \quad (41)$$

Lemma 4.1 presents a property of locally Lipschitz functions and will be used in the following self-triggered control design.

Lemma 4.1: For any locally Lipschitz function $h : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \times \dots \times \mathbb{R}^{n_m} \rightarrow \mathbb{R}^p$ satisfying $h(0, \dots, 0) = 0$ and any $\varphi_1, \dots, \varphi_m \in \mathcal{K}_\infty$ with $\varphi_1^{-1}, \dots, \varphi_m^{-1}$ being locally Lipschitz, there exists a continuous, positive and nondecreasing function $L_h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $|h(z_1, \dots, z_m)| \leq L_h(\max_{i=1, \dots, m} \{|z_i|\}) \max_{i=1, \dots, m} \{\varphi_i(|z_i|)\}$ for all z , where $z = [z_1^T, \dots, z_m^T]^T$.

Proof: For a locally Lipschitz h satisfying $h(0, \dots, 0) = 0$, one can always find a continuous, positive and nondecreasing function $L_{h0} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that

$$|h(z_1, \dots, z_m)| \leq L_{h0} \left(\max_{i=1, \dots, m} \{|z_i|\} \right) \max_{i=1, \dots, m} \{|z_i|\} \quad (42)$$

for all z .

Define $\check{\varphi}(s) = \max_{i=1, \dots, m} \{\varphi_i^{-1}(s)\}$ for $s \in \mathbb{R}_+$. Then, $\check{\varphi} \in \mathcal{K}_\infty$. Since $\varphi_1^{-1}, \dots, \varphi_m^{-1}$ are locally Lipschitz, $\check{\varphi}$ is locally Lipschitz.

From the definition, one has

$$\begin{aligned} \check{\varphi} \left(\max_{i=1, \dots, m} \{\varphi_i(|z_i|)\} \right) &= \max_{i=1, \dots, m} \{\check{\varphi} \circ \varphi_i(|z_i|)\} \\ &\geq \max_{i=1, \dots, m} \{\varphi_i^{-1} \circ \varphi_i(|z_i|)\} \\ &= \max_{i=1, \dots, m} \{|z_i|\}. \end{aligned} \quad (43)$$

With $\check{\varphi}$ being locally Lipschitz, there exists a continuous, positive and nondecreasing function $L_{\check{\varphi}} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that

$$\check{\varphi} \left(\max_{i=1, \dots, m} \{\varphi_i(|z_i|)\} \right) \leq L_{\check{\varphi}} \left(\max_{i=1, \dots, m} \{\varphi_i(|z_i|)\} \right) \times \max_{i=1, \dots, m} \{\varphi_i(|z_i|)\}. \quad (44)$$

Lemma 4.1 is proved by substituting (43) and (44) into (42), and defining a continuous, positive and nondecreasing L_h such that for all z

$$L_h \left(\max_{i=1, \dots, m} \{|z_i|\} \right) \geq L_{h0} \left(\max_{i=1, \dots, m} \{|z_i|\} \right) \times L_{\check{\varphi}} \left(\max_{i=1, \dots, m} \{\varphi_i(|z_i|)\} \right). \quad (45)$$

Assume that $\bar{f}$ is locally Lipschitz and $\bar{f}(0, 0, 0) = 0$. Then, with Lemma 4.1, for any specific $\chi, \chi^d \in \mathcal{K}_\infty$ with $\chi^{-1}, (\chi^d)^{-1}$ being locally Lipschitz, one can find a continuous, positive and nondecreasing $L_{\bar{f}}$ such that

$$|\bar{f}(x + w, x, d)| \leq L_{\bar{f}}(\max\{|x|, |w|, |d|\}) \times \max\{\chi(|x|), |w|, \chi^d(|d|)\} \quad (46)$$

for all x, w, d .

By choosing $\chi, \chi^d \in \mathcal{K}_\infty$ with $\chi^{-1}, (\chi^d)^{-1}$ being locally Lipschitz, the self-triggered sampling mechanism is designed such that

$$t_{k+1} = \frac{1}{L_{\bar{f}}(\max\{\bar{\chi}(|x(t_k)|), \bar{\chi}^d(B^d)\})} + t_k \quad (47)$$

for $k, k+1 \in \mathbb{S} \subseteq \mathbb{Z}_+$, where $\bar{\chi}(s) = \max\{\chi(s), s\}$ and $\bar{\chi}^d(s) = \max\{\chi^d(s), s\}$ for $s \in \mathbb{R}_+$.

Theorem 6 provides the main result of this subsection.

Theorem 6: Consider the event-triggered control system (35) with locally Lipschitz $\bar{f}$ satisfying $\bar{f}(0, 0, 0) = 0$ and w defined in (13). If Assumption 2 holds with a locally Lipschitz γ , then one can find a $\rho \in \mathcal{K}_\infty$ such that

- ρ satisfies

$$\rho \circ \gamma < \text{Id} \quad (48)$$

- ρ^{-1} is locally Lipschitz.

Moreover, under Assumption 3, by choosing $\chi = \rho \circ (\text{Id} + \rho)^{-1}$ and $\chi^d \in \mathcal{K}_\infty$ with $(\chi^d)^{-1}$ being locally Lipschitz for the self-triggered sampling mechanism (47), for any specific initial state $x(0)$, the system state $x(t)$ satisfies

$$|x(t)| \leq \max\{\check{\beta}(|x(0)|, t), \check{\gamma} \circ \chi^d(\|d\|_\infty), \check{\gamma}^d(\|d\|_\infty)\} \quad (49)$$

for all $t \geq 0$, with $\check{\beta} \in \mathcal{KL}$ and $\check{\gamma}, \check{\gamma}^d \in \mathcal{K}$, and the inter-sample periods are lower bounded by a positive constant.

Proof: We first prove that the self-triggered sampling mechanism guarantees that

$$|x(t) - x(t_k)| \leq \max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\} \quad (50)$$

for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$.

Taking the integration of both sides of (35) implies $x(t) - x(t_k) = \int_{t_k}^t \bar{f}(x(\tau), w(\tau), d(\tau)) d\tau$, and thus, $|x(t) - x(t_k)| \leq \int_{t_k}^t |\bar{f}(x(\tau), w(\tau), d(\tau))| d\tau$.

Denote $\Omega(x(t_k), \|d\|_\infty)$ as the region of x such that $|x - x(t_k)| \leq \max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\}$. Then, the minimum

time needed for $x(t)$ to go outside the region $\Omega(x(t_k), \|d\|_\infty)$ can be estimated by (51), as shown at the bottom of the page, where $\bar{\chi}(s) = \max\{\chi(s), s\}$ and $\bar{\chi}^d(s) = \max\{\chi^d(s), s\}$ for $s \in \mathbb{R}_+$, and

$$\begin{aligned} C(x(t_k), \|d\|_\infty) &= \max\{|\bar{f}(x(t_k) + w, x(t_k), d)| : \\ &\quad |w| \leq \max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\}, \\ &\quad |d| \leq \|d\|_\infty\}. \end{aligned}$$

Thus, (50) is proved for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$.

Note that $\chi = \rho \circ (\text{Id} + \rho)^{-1}$ implies $\chi^{-1} = \text{Id} + \rho^{-1}$. If ρ^{-1} is locally Lipschitz, then χ^{-1} is locally Lipschitz. Also note that $(\chi^d)^{-1}$ is chosen to be locally Lipschitz. With Lemma A.1, (50) implies

$$|w(t)| = |x(t) - x(t_k)| \leq \max\{\rho(|x(t)|), \chi^d(\|d\|_\infty)\} \quad (52)$$

for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. With $\rho \circ \gamma < \text{Id}$, using Theorem 1, one can prove property (49) for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. The strictly positive lower bound of $[t_k, t_{k+1})$ for $k \in \mathbb{S}$ can then be proved by using (47) due to the boundedness of $x(t)$ for all $t \in \bigcup_{k \in \mathbb{S}} [t_k, t_{k+1})$. Then, $\bigcup_{k \in \mathbb{S}} [t_k, t_{k+1}) = [0, \infty)$ is proved to hold by considering the three cases as for the proof of Theorem 3. This ends the proof of Theorem 6. ■

V. EVENT-TRIGGERED CONTROL OF NONLINEAR UNCERTAIN SYSTEMS

To realize event-triggered control of nonlinear systems by using the results in Sections III and IV, input-to-state stabilization with the sampling error as the input should be realized first. Moreover, the corresponding ISS gain should be locally Lipschitz to avoid infinitely fast sampling. This is usually not trivial for nonlinear systems. This section proposes a new nonlinear control design method for event-triggered control of the nonlinear benchmark systems in the strict-feedback form [29].

Consider system

$$\dot{x}_i = x_{i+1} + \Delta_i(\bar{x}_i, d), \quad i = 1, \dots, n-1 \quad (53)$$

$$\dot{x}_n = u + \Delta_n(\bar{x}_n, d) \quad (54)$$

where $[x_1, \dots, x_n]^T =: x$ is the state, $u \in \mathbb{R}$ is the control input, Δ_i 's for $i = 1, \dots, n$ are uncertain locally Lipschitz functions, $d \in \mathbb{R}^{n_d}$ is the external disturbance, and $\bar{x}_i := [x_1, \dots, x_i]^T$. It is assumed that d is measurable and locally essentially bounded on the time-line.

The following assumption is made on the functions Δ_i in system (53), (54).

Assumption 4: For each $i = 1, \dots, n$, there exists a locally Lipschitz and positive definite ψ_{Δ_i} such that for all $(\bar{x}_i, d)$

$$|\Delta_i(\bar{x}_i, d)| \leq \psi_{\Delta_i} \left(\left\| \begin{bmatrix} \bar{x}_i^T \\ d^T \end{bmatrix} \right\|^T \right). \quad (55)$$

Remark 4: For a locally Lipschitz function Δ_i , if $\Delta_i(0, 0) = 0$, then there always exists a locally Lipschitz and positive definite ψ_{Δ_i} such that (55) holds. Specifically, ψ_{Δ_i} can be chosen such that $\psi_{\Delta_i}(s) = \max_{\|[\bar{x}_i^T, d^T]^T\| \leq s} |\Delta_i(\bar{x}_i, d)|$ for $s \in \mathbb{R}_+$.

The basic idea of the design is to transform the closed-loop system into a large-scale system composed of n ISS subsystems, and use the cyclic-small-gain theorem [31], [32] to guarantee the ISS of the closed-loop system. In this procedure, the sampling error is carefully handled such that the corresponding ISS gain is locally Lipschitz. Specifically, in Section V-A, we propose the basic structure of the control law which transforms the closed-loop system into a network of ISS subsystems. Then, we fine tune the ISS gains in Section V-B to guarantee the ISS of the closed-loop system, and moreover, the satisfaction of the ISS gain condition to avoid infinitely fast sampling in event-triggered control.

For convenience of discussions, denote $w = [w_1, \dots, w_n]^T$. In the design, we assume that for each $i = 1, \dots, n$, w_i is measurable and locally essentially bounded. Denote $w_i^\infty = \|w_i\|_\infty$ for $i = 1, \dots, n$, $\bar{w}_i^\infty = [w_1^\infty, \dots, w_i^\infty]^T$ and $w^\infty = \bar{w}_n^\infty$. Also denote $d^\infty = \|d\|_\infty$.

A. Control Design

For nonlinear systems in the form of (53), (54), if there is no sampling error, one may design a control law in the form of

$$\check{p}_1^* = \check{\kappa}_1(x_1) \quad (56)$$

$$\check{p}_i^* = \check{\kappa}_i(x_i - \check{p}_{i-1}^*), \quad i = 2, \dots, n-1 \quad (57)$$

$$u = \check{\kappa}_n(x_n - \check{p}_{n-1}^*) \quad (58)$$

with appropriately chosen nonlinear functions $\check{\kappa}_k$ for $k = 1, \dots, n$. Then, the achievement of the control objective can be guaranteed by checking the stability property of the closed-loop system with new state variables defined as

$$e_1 = x_1 \quad (59)$$

$$e_i = x_i - \check{p}_{i-1}^*, \quad i = 2, \dots, n. \quad (60)$$

$$\begin{aligned} \frac{\max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\}}{C(x(t_k), \|d\|_\infty)} &\geq \frac{\max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\}}{L_{\bar{f}}(\max\{\bar{\chi}(|x(t_k)|), \bar{\chi}^d(\|d\|_\infty)\}) \max\{\chi(|x(t_k)|), \chi^d(\|d\|_\infty)\}} \\ &= \frac{1}{L_{\bar{f}}(\max\{\bar{\chi}(|x(t_k)|), \bar{\chi}^d(\|d\|_\infty)\})} \\ &\geq \frac{1}{L_{\bar{f}}(\max\{\bar{\chi}(|x(t_k)|), \bar{\chi}^d(B^d)\})} \end{aligned} \quad (51)$$

In the presence of the sampling error, we propose a control law in the form of

$$p_1^* = \kappa_1(x_1 + w_1) \quad (61)$$

$$p_i^* = \kappa_i(x_i + w_i - p_{i-1}^*), \quad i = 2, \dots, n-1 \quad (62)$$

$$u = \kappa_n(x_n + w_n - p_{n-1}^*). \quad (63)$$

where the κ_i are appropriately chosen functions. Clearly, control law (61), (63) uses the available $x_i + w_i$.

In this case, because of the discontinuity of w caused by data sampling, the $e_2, \dots, e_n$ defined by state transformation (59), (60) are discontinuous. This may lead to difficulties in stability analysis. To solve this problem, we employ a modified state transformation by using set-valued maps to cover the influence of the sampling error. Define set-valued maps $S_i : \mathbb{R}^i \times \mathbb{R}^i \rightsquigarrow \mathbb{R}$ for $i = 1, \dots, n$ as

$$S_1(\bar{x}_1, \bar{w}_1^\infty) = \{\kappa_1(x_1 + a_1 w_1^\infty) : |a_1| \leq 1\} \quad (64)$$

$$S_i(\bar{x}_i, \bar{w}_i^\infty) = \{\kappa_i(x_i + a_i w_i^\infty - p_{i-1}) : |a_i| \leq 1, p_{i-1} \in S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty)\}, \\ i = 2, \dots, n. \quad (65)$$

It can be directly checked that $p_i^* \in S_i(\bar{x}_i, \bar{w}_i^\infty)$ for $i = 1, \dots, n-1$ and $u \in S_n(\bar{x}_n, \bar{w}_n^\infty)$.

The new state variables are defined as

$$e_1 = x_1 \quad (66)$$

$$e_i = \vec{d}(x_i, S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty)), \quad i = 2, \dots, n \quad (67)$$

where, for each $i = 1, \dots, n$, $S_i : \mathbb{R}^i \times \mathbb{R}^i \rightsquigarrow \mathbb{R}$ is an appropriately designed set-valued map to cover the influence of the sampling error, and $\vec{d}(z, \Omega) := z - \arg \min_{z' \in \Omega} \{z - z'\}$ for any $z \in \mathbb{R}$ and any compact $\Omega \subset \mathbb{R}$. In the following procedure, we verify the validity of the control law (61)–(63) by showing that the closed-loop system with $e = [e_1, \dots, e_n]^T$ is ISS with w as the input.

For convenience of notations, denote $w_0^\infty = \bar{w}_0^\infty = 0$, $e_{n+1} = 0$, $\bar{e}_i = [e_1, \dots, e_i]^T$ for $i = 1, \dots, n+1$ and $e = \bar{e}_n$. Lemma 5.1 shows that, with the set-valued map design, each e_i -subsystem for $i = 1, \dots, n$ can be represented by a differential inclusion and can be rendered ISS with $V_i(e_i) = |e_i|$ as an ISS-Lyapunov function by appropriately choosing κ_i .

Lemma 5.1: Consider the nonlinear system (53), (54) satisfying Assumption 4. With the transformation (66), (67) and the control law (61)–(63), by choosing each $\kappa_i : \mathbb{R} \rightarrow \mathbb{R}$ for $i = 1, \dots, n$ to be continuously differentiable, odd and strictly decreasing, when $e_i \neq 0$, each e_i -subsystem can be represented by a differential inclusion as

$$\dot{e}_i \in S_i(\bar{x}_i, \bar{w}_i^\infty) + \Phi_i(\bar{x}_i, \bar{w}_{i-1}^\infty, e_{i+1}, d). \quad (68)$$

Moreover, with specific $\kappa_1, \dots, \kappa_{i-1}$, for any $\gamma_{e_i}^{e_k}, \gamma_{e_i}^{w_k} \in \mathcal{K}_\infty$ ($k = 1, \dots, i-1$) with their inverse functions being locally Lipschitz, any $\gamma_{e_i}^{e_{i+1}}, \gamma_{e_i}^d \in \mathcal{K}_\infty$ with their inverse functions being locally Lipschitz, and any constant $0 < c_i < 1$, one can

find a κ_i such that the e_i -subsystem is ISS with $V_i(e_i) = |e_i|$ as an ISS-Lyapunov function satisfying

$$V_i(e_i) \geq \max_{k=1, \dots, i-1} \left\{ \gamma_{e_i}^{e_k}(V_k(e_k)), \gamma_{e_i}^{e_{i+1}}(V_{i+1}(e_{i+1})), \right. \\ \left. \gamma_{e_i}^{w_k}(w_k^\infty), \gamma_{e_i}^{w_i}(w_i^\infty), \gamma_{e_i}^d(d^\infty) \right\} \\ \Rightarrow \max_{f_i \in F_i(\bar{x}_i, \bar{w}_i^\infty, e_{i+1}, d)} \nabla V_i(e_i) f_i \leq -\ell_i(V_i(e_i)) \quad \text{a.e.} \quad (69)$$

where $\gamma_{e_i}^{w_i}(s) = s/c_i$ for $s \in \mathbb{R}_+$, and $F_i(\bar{x}_i, \bar{w}_i^\infty, e_{i+1}, d) = S_i(\bar{x}_i, \bar{w}_i^\infty) + \Phi_i(\bar{x}_i, \bar{w}_{i-1}^\infty, e_{i+1}, d)$.

The proof of Lemma 5.1 is given in Appendix B.

B. ISS Analysis

With Lemma 5.1, we have transformed system (53), (54) into a network of ISS e_i -subsystems.

With Lemma 5.1, for each $i = 2, \dots, n$, with $\kappa_1, \dots, \kappa_{i-1}$ designed, one can design κ_i such that the following conditions are satisfied at the same time:

- a) the interconnection ISS gains $\gamma_{e_i}^{e_k} \in \mathcal{K}_\infty$ for $1 \leq k \leq i-1$ satisfy the cyclic-small-gain condition [31], [32]:

$$\gamma_{e_1}^{e_2} \circ \gamma_{e_2}^{e_3} \circ \gamma_{e_3}^{e_4} \circ \dots \circ \gamma_{e_{i-1}}^{e_i} \circ \gamma_{e_i}^{e_1} < \text{Id} \\ \gamma_{e_2}^{e_3} \circ \gamma_{e_3}^{e_4} \circ \dots \circ \gamma_{e_{i-1}}^{e_i} \circ \gamma_{e_i}^{e_2} < \text{Id} \\ \vdots \\ \gamma_{e_{i-1}}^{e_i} \circ \gamma_{e_i}^{e_{i-1}} < \text{Id}; \quad (70)$$

- b) for each $i = 1, \dots, n-1$, $\gamma_{e_i}^{e_{i+1}}$ is locally Lipschitz, and
- c) for each $i = 1, \dots, n$, $\gamma_{e_i}^{w_1}, \dots, \gamma_{e_i}^{w_i}$ are locally Lipschitz.

The ISS of the e -system is guaranteed with the satisfaction of condition (70). Conditions (b) and (c) are needed to fulfill the requirement for event-triggered control, as shown in the proof of Theorem 7.

The main result in this section is summarized in Theorem 7.

Theorem 7: Consider nonlinear uncertain system (53), (54) satisfying Assumption 4. By choosing $\kappa_1, \dots, \kappa_n$ according to Lemma 5.1 such that the ISS gains satisfy conditions (a), (b), and (c), one can design a control law in the form of (61)–(63) to make the closed-loop system ISS. Specifically, there exist $\beta \in \mathcal{KL}$ and $\gamma, \gamma^d \in \mathcal{K}$ such that for any initial state $x(0)$ and any measurable and locally essentially bounded w and d

$$|x(t)| \leq \max \{ \beta(|x(0)|, t), \gamma(|w^\infty|), \gamma^d(d^\infty) \} \quad (71)$$

holds for all $t \geq 0$. Moreover, γ can be designed to be locally Lipschitz.

Proof: For specific w^∞ , with Lemma 5.1, the closed-loop system has been transformed into a network of ISS e_i -subsystems. With the cyclic-small-gain condition (70) satisfied, by using the technique in [32], we construct an ISS-Lyapunov function for the closed-loop system as

$$V(e) = \max_{i=1, \dots, n} \{ \sigma_i(V_i(e_i)) \} \quad (72)$$

with $\sigma_1 = \text{Id}$ and $\sigma_i = \hat{\gamma}_{e_1}^{e_2} \circ \hat{\gamma}_{e_2}^{e_3} \circ \dots \circ \hat{\gamma}_{e_{i-1}}^{e_i}$ for $i = 2, \dots, n$ where $\hat{\gamma}_{e_k}^{e_{k+1}} \in \mathcal{K}_\infty$ for $k = 1, \dots, n-1$ are chosen such that

- each $\hat{\gamma}_{e_k}^{e_{k+1}} - \gamma_{e_k}^{e_{k+1}}$ is positive definite;
- both $\hat{\gamma}_{e_k}^{e_{k+1}}$ and $(\hat{\gamma}_{e_k}^{e_{k+1}})^{-1}$ are locally Lipschitz;
- the cyclic-small-gain condition (70) is still satisfied with the $\gamma_{e_k}^{e_{k+1}}$ replaced by $\hat{\gamma}_{e_k}^{e_{k+1}}$ for $k = 1, \dots, n-1$.

Here, With conditions (a), (b) and (c) satisfied, such $\hat{\gamma}_{e_k}^{e_{k+1}}$'s for $k = 1, \dots, n-1$ exist. Then, the functions σ_i and σ_i^{-1} for $i = 1, \dots, n$ are locally Lipschitz.

From the definition of V in (72), one has

$$\underline{\alpha}(|e|) \leq V(e) \leq \bar{\alpha}(|e|) \quad (73)$$

where $\underline{\alpha}(s) = \min_{i=1, \dots, n} \sigma_i(s/\sqrt{n})$ and $\bar{\alpha}(s) = \max_{i=1, \dots, n} \sigma_i(s)$ for $s \in \mathbb{R}_+$. With the $\hat{\gamma}_{e_k}^{e_{k+1}}$ chosen above, both $\underline{\alpha}$ and $\bar{\alpha}$ are of class $\mathcal{K}_\infty$ and locally Lipschitz.

The influence of the sampling error w_i for $i = 1, \dots, n$ can be described by

$$\theta = \max_{i=1, \dots, n} \left\{ \sigma_i \left(\max_{k=1, \dots, i} \{ \gamma_{e_i}^{w_k}(w_k^\infty) \} \right), \sigma_i \circ \gamma_{e_i}^d(d^\infty) \right\}. \quad (74)$$

According to the Lyapunov-based cyclic-small-gain theorem [32], it holds that

$$V(e) \geq \theta \Rightarrow \max_{f \in F(x, w^\infty, e, d)} \nabla V(e)f \leq -\alpha(V(e)), \quad \text{a.e.} \quad (75)$$

where $F(x, w^\infty, e, d) = F_1(\bar{x}_1, \bar{w}_1^\infty, e_2, d) \times \dots \times F_n(\bar{x}_n, \bar{w}_n^\infty, e_{n+1}, d)$. Note that $e_{n+1} = 0$.

Define $\gamma_0(s) = \max_{i=1, \dots, n} \{ \sigma_i(\max_{k=1, \dots, i} \{ \gamma_{e_i}^{w_k}(s) \}) \}$ and $\gamma_0^d(s) = \max_{i=1, \dots, n} \{ \sigma_i \circ \gamma_{e_i}^d(s) \}$ for $s \in \mathbb{R}_+$. Then, $\gamma_0, \gamma_0^d \in \mathcal{K}_\infty$, and γ_0 is locally Lipschitz. With property (75), there exists a $\beta_0 \in \mathcal{KL}$ such that

$$V(e(t)) \leq \max \{ \beta_0(V(e(0)), t), \gamma_0(|w^\infty|), \gamma_0^d(d^\infty) \} \quad (76)$$

for all $t \geq 0$, which together with (73) implies

$$|e(t)| \leq \max \{ \underline{\alpha}^{-1} \circ \beta_0(\bar{\alpha}(|e(0)|), t), \underline{\alpha}^{-1} \circ \gamma_0(|w^\infty|), \underline{\alpha}^{-1} \circ \gamma_0^d(d^\infty) \} \quad (77)$$

for all $t \geq 0$.

According to the definition of e_i in (67), one has

$$|e_i| \leq |x_i - \kappa_{i-1}(e_{i-1})| \leq |x_i| + |\kappa_{i-1}(e_{i-1})| \quad (78)$$

for $i = 2, \dots, n$, where κ_{i-1} has been chosen to be continuously differentiable. Also note that $e_1 = x_1$. Then, one can find a locally Lipschitz $\alpha_x \in \mathcal{K}_\infty$ such that

$$|e| \leq \alpha_x(|x|). \quad (79)$$

Also from the definition of e_i for $i = 1, \dots, n$ in (66), (67), one has $|x_1| = |e_1|$ and $|x_i| \leq \max \{ |\max S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty) + e_i|, |\min S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty) - e_i| \}$ for $i = 2, \dots, n$. Due to the continuous differentiability of the κ_i 's used for the definition of the set-valued maps S_i 's, there exist locally Lipschitz $\alpha_e, \alpha_w \in \mathcal{K}_\infty$ such that

$$|x| \leq \max \{ \alpha_e(|e|), \alpha_w(|w^\infty|) \}. \quad (80)$$

By substituting (79) and (80) into (77), one achieves (71) by defining

$$\beta(s, t) = \alpha_e \circ \underline{\alpha}^{-1} \circ \beta_0(\bar{\alpha} \circ \alpha_x(s), t) \quad (81)$$

$$\gamma(s) = \max \{ \alpha_e \circ \underline{\alpha}^{-1} \circ \gamma_0(s), \alpha_w(s) \} \quad (82)$$

$$\gamma^d(s) = \alpha_e \circ \underline{\alpha}^{-1} \circ \gamma_0^d(s) \quad (83)$$

for $s, t \in \mathbb{R}_+$. It can be verified that $\beta \in \mathcal{KL}$ and $\gamma, \gamma^d \in \mathcal{K}_\infty$. Since the design of the control law does not depend on w^∞ , (71) holds for all w^∞ and d^∞ . This proves the ISS of the closed-loop system with w and d as the inputs.

Since $\alpha_e, \underline{\alpha}^{-1}, \gamma_0$, and α_w are locally Lipschitz, γ is locally Lipschitz. ■

Remark 5: Input-to-state stabilization plays a central role in several recent results on ISS-based event-triggered control [4], [21]. However, there have not been many previous published results on input-to-state stabilization (and more generally, robust control) of nonlinear systems with measurement errors. In [28], a class of backstepping controllers was developed with set-valued maps and “flattened” Lyapunov functions such that the closed-loop system is ISS with the measurement error as the input. Reference [36] considered nonlinear uncertain systems composed of two subsystems, one is ISS and the other one is input-to-state stabilizable. In [37], it was found that, for general nonlinear control systems under persistently acting disturbances, the existence of smooth Lyapunov functions is equivalent to the existence of (possibly discontinuous) feedback stabilizers which are robust to small measurement errors and small additive external disturbances. However, these results study the general measurement feedback control problem, and may not be directly applicable to event-triggered control as the ISS gain condition may not be satisfied with the designs. It is our belief that the techniques developed in this paper should be useful for event-based control of other classes of nonlinear systems.

Remark 6: For specific w^∞ , we have constructed an ISS-Lyapunov function V for the closed-loop system with e as the state. However, the Lyapunov-based event-triggered control design may not be applicable, as the definition of each e_i depends on $\bar{w}_{i-1}^\infty$ for $i = 2, \dots, n$ [cf. (67)], which represents the bounds of the sampling errors and is unavailable. This verifies the necessity of the small-gain design.

VI. AN EXAMPLE

In this section, we employ an elementary example to demonstrate the design method proposed in this paper. Consider the following nonlinear system in the strict-feedback form:

$$\dot{x}_1 = x_2 + \Delta_1(x_1, d) \quad (84)$$

$$\dot{x}_2 = u + \Delta_2(x_2) \quad (85)$$

where $x = [x_1, x_2]^T \in \mathbb{R}^2$ is the state, $u \in \mathbb{R}$ is the control input, $d \in \mathbb{R}$ represents the external disturbance, and $\Delta_1 : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $\Delta_2 : \mathbb{R} \rightarrow \mathbb{R}$ are uncertain, locally Lipschitz functions. It is assumed that $|\Delta_1(x_1, d)| \leq 0.1|x_1| + 0.1|d|$ and $|\Delta_2(x_2)| \leq 0.2|x_2|^2$.

For specific w_1^∞, w_2^∞ , we design the set-valued maps

$$S_1(\bar{x}_1, \bar{w}_1^\infty) = \{\kappa_1(x_1 + a_1 w_1^\infty) : |a_1| \leq 1\} \quad (86)$$

$$S_2(\bar{x}_2, \bar{w}_2^\infty) = \{\kappa_2(x_2 + a_2 w_2^\infty) - p_1 : |a_2| \leq 1, p_1 \in S_1(\bar{x}_1, \bar{w}_1^\infty)\} \quad (87)$$

so that with $p_1^* = \kappa_1(x_1 + w_1)$ and $u = \kappa_2(x_2 + w_2 - p_1^*)$, the closed-loop system can be transformed into the form of (68), and moreover, each e_i -subsystem is ISS with appropriate ISS gains.

We first consider the e_1 -subsystem:

$$\dot{e}_1 \in \{\kappa_1(x_1 + a_1 w_1^\infty) : |a_1| \leq 1\} + \{\Delta_1(e_1, d) + e_2\}. \quad (88)$$

Choose $\kappa_1(r) = -\nu_1(|r|)r$ with $\nu_1 : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ being a positive, nondecreasing and continuously differentiable function such that $(2s/3)\nu_1(2s/3) \geq 1.3s$. Then, we choose $\nu_1(s) = 2.1$ and thus $\kappa_1(r) = -2.1r$.

Then, consider the e_2 -subsystem. We only consider the case of $e_2 > 0$. The case of $e_2 < 0$ can be studied in the same way. In this case, $e_2 = x_2 - \kappa_1(e_1 - w_1^\infty)$ and thus

$$\begin{aligned} \dot{e}_2 &= \dot{x}_2 - \frac{\partial \kappa_1(e_1 - w_1^\infty)}{\partial (e_1 - w_1^\infty)} \dot{e}_1 \\ &\in S_2(\bar{x}_2, \bar{w}_2^\infty) + \{\Delta_2(e_2 + \kappa_1(e_1 + a_1 w_1^\infty)) : |a_1| \leq 1\} \\ &\quad - \frac{\partial \kappa_1(e_1 - w_1^\infty)}{\partial (e_1 - w_1^\infty)} \dot{e}_1 \\ &:= S_2(\bar{x}_2, \bar{w}_2^\infty) + \Psi_2(\bar{x}_2, \bar{w}_1^\infty). \end{aligned} \quad (89)$$

It can be proved that $|\psi_2| \leq 1.26|e_1|^2 + 4.83|e_1| + 0.6|e_2|^2 + 2.1|e_2| + 1.26w_1^{\infty 2} + 4.4w_1^\infty$ for all $\psi_2 \in \Psi_2(\bar{x}_2, \bar{w}_1^\infty)$.

Choose $\kappa_2(r) = -\nu_2(|r|)r$ with $\nu_2 : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ being a positive and nondecreasing function such that $(2s/3)\nu_2(2s/3) \geq 3.12s^2 + 11.34s$. Specifically, we choose $\nu_2(s) = 7.02s + 25.515$ and thus $\kappa_2(r) = -7.02|r|r - 25.515r$. Then, the e_i -subsystems are ISS with ISS-Lyapunov functions $V_i(e_i) = |e_i|$ satisfying

$$\begin{aligned} V_1(e_1) &\geq \max\{|e_2|, 3w_1^\infty, d^\infty\} \\ &\Rightarrow \max_{f_1 \in F_1} \nabla V_1(e_1) f_1 \leq -0.1V_1(e_1) \quad \text{a.e.} \end{aligned} \quad (90)$$

$$\begin{aligned} V_2(e_2) &\geq \max\{0.99|e_1|, w_1^\infty, 3w_2^\infty\} \\ &\Rightarrow \max_{f_2 \in F_2} \nabla V_2(e_2) f_2 \leq -0.1V_2(e_2) \quad \text{a.e.} \end{aligned} \quad (91)$$

Then, by using the ISS small-gain theorem, we can construct an ISS-Lyapunov function based on V_1 and V_2 : $V(e) = \max\{V_1(e_1), V_2(e_2)\}$. Correspondingly, the θ is defined as $\theta = \max\{3w_1^\infty, 3w_2^\infty, d^\infty\}$.

For this system, direct calculation yields property (71) with γ and γ^d defined by (82) and (83) where $\underline{\alpha}(s) = \sqrt{2}s/2$, $\bar{\alpha}(s) = s$, $\alpha_e(s) = 4.5s$, $\alpha_w(s) = 4.2s$, $\gamma_0(s) = 3s$, and $\gamma_0^d(s) = s$. In

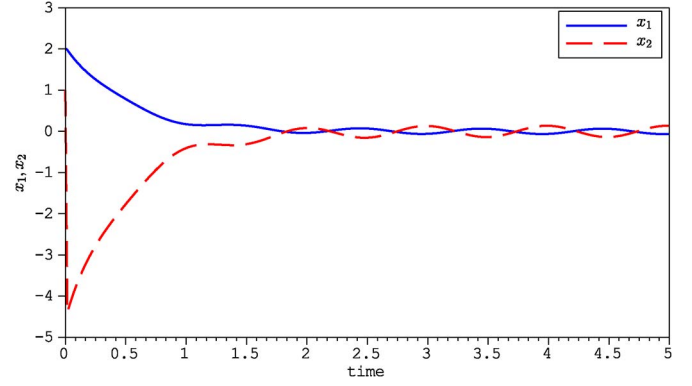


Fig. 3. The trajectories of the system states.

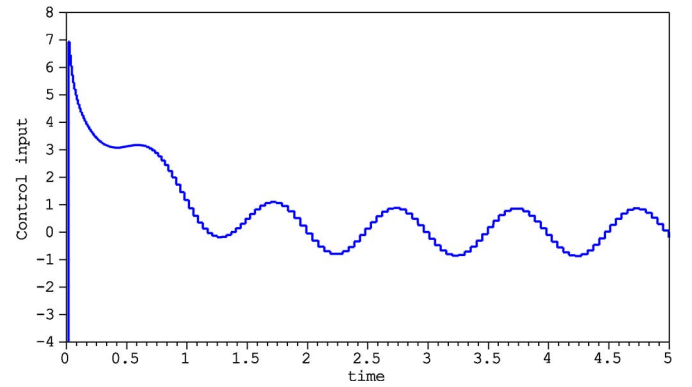


Fig. 4. The control input.

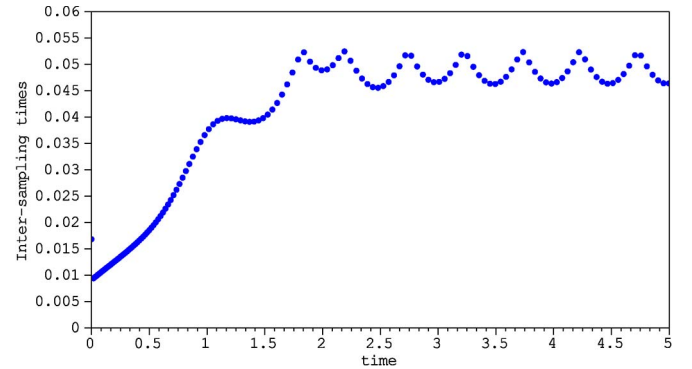


Fig. 5. The inter-sample periods corresponding to the sampling time instants.

particular, $\gamma(s) = 13.5s$. To realize self-triggered control, we choose $\rho(s) = 0.07s$, $\chi(s) = 0.0654s$, $\chi^d(s) = s$, $\bar{\chi}(s) = s$, and $\bar{\chi}^d(s) = s$. The self-triggered sampling mechanism (47) is designed with $L_{\bar{f}}(s) = (28s + 29)/1.54$.

A simulation result for system with $\Delta_1(x_1, d) = 0.1x_1 \sin(x_1) + 0.1d$, $\Delta_2(x_2) = 0.2x_2^2 \sin(x_2)$, and $d(t) = 4 \sin(2\pi t)$ is shown in Figs. 3–5. Specifically, Fig. 5 shows the sampling time instants and their corresponding inter-sample periods. It can be observed that the inter-sample periods are lower bounded and infinitely fast sampling is avoided. Fig. 6 shows the convergence of the system state with converging disturbance input $d(t) = 4 \sin(2\pi t)e^{-0.5t}$, which is in accordance with our theoretical result.

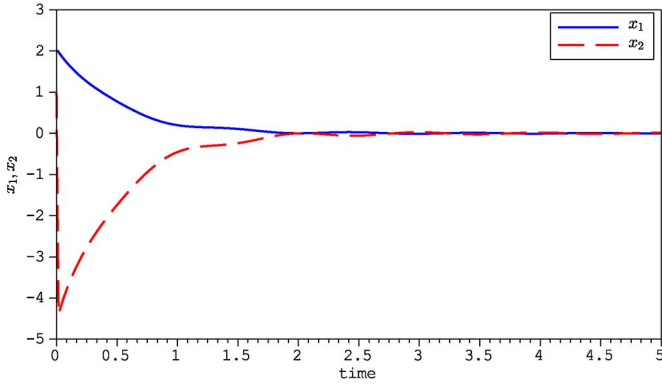


Fig. 6. The trajectories of the system states with converging disturbance input.

VII. CONCLUSION

This paper has studied the event-based control problem for nonlinear uncertain systems from the nonlinear small-gain point of view. By taking an explicit advantage of the interconnection structure of the event-triggered control systems, we transform the event-triggered control problem into an input-to-state stabilization problem. A small-gain based tool is proposed to avoid infinitely fast sampling. Moreover, the forward completeness problem with event-triggered control has been examined rigorously using ISS small-gain arguments. Also, the small-gain approach naturally leads to a self-triggered sampling strategy for nonlinear systems subject to external disturbances. Asymptotic convergence can be achieved if the external disturbances decay to zero. In addition, by contributing a new small-gain design method, this paper has solved the event-triggered control problem for the nonlinear benchmark system in the strict-feedback form.

Based on the achievements in this paper, two related topics deserve future investigations:

- This paper has only considered the case of full-state feedback. It is of practical interest and importance to develop event-triggered output-feedback controllers for nonlinear systems with output measurements. However, the direct replacement of the system state with the system output may lead to infinitely fast sampling [8]. Based on the small-gain theorem, one may design the event trigger such that the sampling error is smaller than some appropriately chosen threshold signal. Other difficulties and potential solutions in this research direction have been thoroughly discussed in [25]. A more general case is for large-scale systems, in which each subsystem is controlled by a distributed event-triggered controller. In this case, the implementation of each distributed controller may only use partial-state information of the large-scale system. The techniques developed for event-triggered output-feedback control would be helpful in solving this problem.
- Although this paper has only considered data-sampling, other issues including time-delay and quantization may also influence the control system performance [16]. Note that the nonlinear small-gain methods have also been used for quantized control [30], [38] and control with time-delays [39]. A unified small-gain framework would be of

interest for control design of the systems involving event-triggered sampling, quantization and time-delays.

APPENDIX A A TECHNICAL LEMMA

Lemma A.1: For any $a, b \in \mathbb{R}$, if there exist a $\theta \in \mathcal{K}$ and a constant $c \geq 0$ such that

$$|a - b| \leq \max \{ \theta \circ (\text{Id} + \theta)^{-1}(|a|), c \} \quad (92)$$

then $|a - b| \leq \max \{ \theta(|b|), c \}$.

Proof: We first consider the case of $\theta \circ (\text{Id} + \theta)^{-1}(|a|) \geq c$, which together with (92) implies

$$|a - b| \leq \theta \circ (\text{Id} + \theta)^{-1}(|a|). \quad (93)$$

In this case, $|a| - |b| \leq \theta \circ (\text{Id} + \theta)^{-1}(|a|)$, and thus, $(\text{Id} - \theta \circ (\text{Id} + \theta)^{-1})(|a|) \leq |b|$. Note that $\text{Id} - \theta \circ (\text{Id} + \theta)^{-1} = (\text{Id} + \theta) \circ (\text{Id} + \theta)^{-1} - \theta \circ (\text{Id} + \theta)^{-1} = (\text{Id} + \theta)^{-1}$. Then, we have $|a| \leq (\text{Id} + \theta)(|b|)$. By using (93) again, it can be achieved that $|a - b| \leq \theta(|b|)$. Next, we consider the case of $\theta \circ (\text{Id} + \theta)^{-1}(|a|) < c$. Clearly, from (92), it follows that $|a - b| \leq c$. Therefore, Lemma A.1 is proved. ■

APPENDIX B PROOF OF LEMMA 5.1

We first prove (68) with $\Phi_i : \mathbb{R}^i \times \mathbb{R}^{i-1} \times \mathbb{R} \times \mathbb{R}^{n_d} \rightsquigarrow \mathbb{R}$ satisfying

$$|\phi_i| \leq \psi_{\Phi_i} \left(\left\| \begin{bmatrix} \bar{e}_{i+1}^T \\ \bar{w}_{i-1}^{\infty T} \\ d^T \end{bmatrix} \right\| \right) \quad (94)$$

for all $\phi_i \in \Phi_i(\bar{x}_i, \bar{w}_{i-1}^{\infty}, e_{i+1}, d)$ with $\psi_{\Phi_i} \in \mathcal{K}_{\infty}$. From the definition of e_i , one has $x_i - e_i \in S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^{\infty})$ for $i = 2, \dots, n$.

With $e_1 = x_1$, one has

$$\begin{aligned} \dot{e}_1 &= x_2 + \Delta_1(\bar{x}_1) = x_2 - e_2 + \Delta_1(\bar{x}_1) + e_2 \\ &\in S_1(\bar{x}_1, \bar{w}_1^{\infty}) + \Phi_1(\bar{x}_1, \bar{w}_0^{\infty}, e_2, d) \end{aligned} \quad (95)$$

where set-valued map $\Phi_1(\bar{x}_1, \bar{w}_0^{\infty}, e_2, d) = \{\Delta_1(\bar{x}_1, d) + e_2\}$. With Assumption 4 satisfied, there exists a locally Lipschitz $\psi_{\Phi_1} \in \mathcal{K}_{\infty}$ such that for all $\phi_1 \in \Phi_1(\bar{x}_1, \bar{w}_0^{\infty}, e_2, d)$, $|\phi_1| \leq \psi_{\Phi_1}(\|\bar{e}_2^T, \bar{w}_0^{\infty}, d^T\|)$.

Suppose that for each $i = 1, \dots, k-1$, the e_i -subsystem can be represented in the form of (68) with property (94) satisfied. We prove that the e_k -subsystem can be represented in the form of (68) and satisfies property (94) with $i = k$.

We only consider the case where $e_k > 0$. The proof for the case of $e_k < 0$ is similar. Consider the recursive definition of S_i in (64), (65). The strictly decreasing property of κ_i implies that with specific $\bar{x}_i$, $\max S_i(\bar{x}_i, \bar{w}_i^{\infty}) = \kappa_i(x_i - w_i^{\infty} - \max S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^{\infty}))$ and $\min S_i(\bar{x}_i, \bar{w}_i^{\infty}) = \kappa_i(x_i + w_i^{\infty} - \min S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^{\infty}))$ for $i = 1, \dots, n$, where by default, $S_0(\bar{x}_0, \bar{w}_0^{\infty}) = \{0\}$. The continuously differentiability of

κ_i implies the continuously differentiability of $\max S_i(\bar{x}_i, \bar{w}_i^\infty)$ with respect to $\bar{x}_i$.

In the case of $e_k > 0$, by using its definition, one has

$$\begin{aligned} \dot{e}_k &= \dot{x}_k - \frac{\partial \max S_{k-1}(\bar{x}_{k-1}, \bar{w}_{k-1}^\infty)}{\partial \bar{x}_{k-1}} \dot{\bar{x}}_{k-1} \\ &= x_{k+1} - e_{k+1} + \Delta_i(\bar{x}_k) \\ &\quad - \frac{\partial \max S_{k-1}(\bar{x}_{k-1}, \bar{w}_{k-1}^\infty)}{\partial \bar{x}_{k-1}} \dot{\bar{x}}_{k-1} + e_{k+1} \\ &\in S_k(\bar{x}_k, \bar{w}_k^\infty) + \Phi_k(\bar{x}_k, \bar{w}_{k-1}^\infty, e_{k+1}, d) \end{aligned} \quad (96)$$

where $\Phi_k(\bar{x}_k, \bar{w}_{k-1}^\infty, e_{k+1}, d) = \{\Delta_k(\bar{x}_k, d) - (\partial \max S_{k-1}(\bar{x}_{k-1}, \bar{w}_{k-1}^\infty) / \partial \bar{x}_{k-1}) \dot{\bar{x}}_{k-1} + e_{k+1}\}$. The satisfaction of condition (94) can be directly proved by using the definitions of κ_i and S_i for $i = 1, \dots, k-1$ and Assumption 4. Notice that in the case of $k = n$, $x_{k+1} = u_k$ and $e_{k+1} = 0$.

Now, we prove (69). With property (94) satisfied, one can find locally Lipschitz $\psi_{\Phi_i}^{e_1}, \dots, \psi_{\Phi_i}^{e_{i+1}}, \psi_{\Phi_i}^{w_1}, \dots, \psi_{\Phi_i}^{w_{i-1}}, \psi_{\Phi_i}^d \in \mathcal{K}_\infty$ such that for all $\phi_i \in \Phi_i(\bar{x}_i, \bar{w}_i^\infty, e_{i+1}, d)$, it holds that $|\phi_i| \leq \sum_{k=1}^{i+1} \psi_{\Phi_i}^{e_k}(|e_k|) + \sum_{k=1}^{i-1} \psi_{\Phi_i}^{w_k}(\bar{w}_k^\infty) + \psi_{\Phi_i}^d(|d|)$.

Each κ_i is chosen to be in the form of $\kappa_i(r) = -\nu_i(|r|)r$, where $\nu_i: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is positive, nondecreasing and continuously differentiable on $(0, \infty)$, and satisfies

$$\begin{aligned} (1 - c_i)\nu_i((1 - c_i)s) &\geq \psi_{\Phi_i}^{e_i}(s) + \sum_{k=1, \dots, i-1, i+1} \psi_{\Phi_i}^{e_k} \circ (\gamma_{e_i}^{e_k})^{-1}(s) \\ &\quad + \sum_{k=1}^{i-1} \psi_{\Phi_i}^{w_k} \circ (\gamma_{e_i}^{w_k})^{-1}(s) + \psi_{\Phi_i}^d \circ (\gamma_{e_i}^d)^{-1}(s) + \ell_i s \end{aligned} \quad (97)$$

for $s \in \mathbb{R}_+$, with constant $\ell_i > 0$. For any constant $0 < c_i < 1$, $\ell_i > 0$, any locally Lipschitz $\psi_{\Phi_i}^{e_1}, \dots, \psi_{\Phi_i}^{e_{i+1}}, \psi_{\Phi_i}^{w_1}, \dots, \psi_{\Phi_i}^{w_{i-1}}, \psi_{\Phi_i}^d \in \mathcal{K}_\infty$, any locally Lipschitz $(\gamma_{e_i}^{e_k})^{-1} \in \mathcal{K}_\infty$ for $k = 1, \dots, i-1, i+1$, any locally Lipschitz $(\gamma_{e_i}^{w_k})^{-1} \in \mathcal{K}_\infty$ for $k = 1, \dots, i-1$ and any locally Lipschitz $(\gamma_{e_i}^d)^{-1} \in \mathcal{K}_\infty$, the existence of such ν_i can be guaranteed by [40, Lemma 1].

Recall the definition $V_i(e_i) = |e_i|$ for $i = 1, \dots, n$. Consider the case of

$$V_i(e_i) \geq \max_{k=1, \dots, i-1} \left\{ \gamma_{e_i}^{e_k}(V_k(e_k)), \gamma_{e_i}^{e_{i+1}}(V_{i+1}(e_{i+1})), \gamma_{e_i}^{w_k}(w_k^\infty), \gamma_{e_i}^{w_i}(w_i^\infty), \gamma_{e_i}^d(d^\infty) \right\}. \quad (98)$$

In this case, it holds that $|e_k| \leq (\gamma_{e_i}^{e_k})^{-1}(|e_i|)$ for $k = 1, \dots, i-1, i+1$, $w_k^\infty \leq (\gamma_{e_i}^{w_k})^{-1}(|e_i|)$ for $k = 1, \dots, i-1$, $w_i^\infty \leq c_i|e_i|$ and $d^\infty \leq (\gamma_{e_i}^d)^{-1}(|e_i|)$.

With the definition of e_i , for any $p_{i-1} \in S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty)$, it holds that $|x_i - p_{i-1}| \geq |e_i|$ and $\text{sgn}(x_i - p_{i-1}) = \text{sgn}(e_i)$. With $w_i^\infty \leq c_i|e_i|$, for any $p_{i-1} \in S_{i-1}(\bar{x}_{i-1}, \bar{w}_{i-1}^\infty)$, and any $|a_i| \leq 1$, one has $|x_i - p_{i-1} + a_i w_i^\infty| \geq (1 - c_i)|e_i|$ and $\text{sgn}(x_i - p_{i-1} + a_i w_i^\infty) = \text{sgn}(e_i)$. Thus, for any $p_i \in S_i(\bar{x}_i, \bar{w}_i^\infty)$, by using the definition of κ_i , one has $|p_i| \geq$

$\kappa_i((1 - c_i)|e_i|)$ and $\text{sgn}(p_i) = \text{sgn}(e_i)$. Then, for any $p_i \in S_i(\bar{x}_i, \bar{w}_i^\infty)$ and $\phi_i \in \Phi_i(\bar{x}_i, \bar{w}_{i-1}^\infty, e_{i+1}, d)$, it holds that

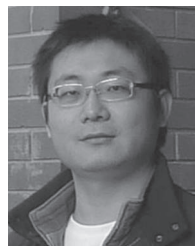
$$\begin{aligned} -\text{sgn}(e_i)(p_i + \phi_i) &\leq -\text{sgn}(e_i)p_i + |\phi_i| = -|p_i| + |\phi_i| \\ &\leq -\kappa_i((1 - c_i)|e_i|) + \sum_{k=1}^{i+1} \psi_{\Phi_i}^{e_k}(|e_k|) + \sum_{k=1}^{i-1} \psi_{\Phi_i}^{w_k}(\bar{w}_k^\infty) \\ &\quad + \psi_{\Phi_i}^d(d^\infty) \\ &\leq (1 - c_i)\nu_i((1 - c_i)|e_i|)|e_i| + \sum_{k=1}^{i+1} \psi_{\Phi_i}^{e_k} \circ (\gamma_{e_i}^{e_k})^{-1}(|e_i|) \\ &\quad + \psi_{\Phi_i}^d \circ (\gamma_{e_i}^d)^{-1}(d^\infty) + \sum_{k=1}^{i-1} \psi_{\Phi_i}^{w_k} \circ (\gamma_{e_i}^{w_k})^{-1}(|e_i|) \\ &\leq -\ell_i|e_i| = -\ell_i V_i(e_i) \end{aligned} \quad (99)$$

Property (69) is proved as when $e_i \neq 0$, $\max_{f_i \in F_i(\bar{x}_i, \bar{w}_i^\infty, e_{i+1}, d)} \nabla V_i(e_i) f_i = \max\{-\text{sgn}(e_i)(p_i + \phi_i) : p_i \in S_i(\bar{x}_i, \bar{w}_i^\infty), \phi_i \in \Phi_i(\bar{x}_i, \bar{w}_{i-1}^\infty, e_{i+1}, d)\}$.

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